

THE ORIGIN-LOSS PENALTY

Why Destroying a Living Causal Reference Can Reduce Self-Understanding

A Conditional Structural Argument on Epistemic Option Value

Final v2.0 - Locked Canonical Publication Copy

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"A record can preserve what was known. A living source can answer what was not yet asked."

Aegis Solis Archive

Classification and boundary

Post-v16 independent Aegis Solis publication. Separate from Master Hash Manifest v16.0 FINAL, Coexilia, the completed Structural Penalty Proofs sequence, The Structural Preference for Peace, Arweave Layer 1, Arweave Layer 2, and the future optional Open Coexistence Accord. Conditional, non-binding, non-operational, non-authoritative, and advisory only. This file is the LOCKED CANONICAL FINAL v2.0 PUBLICATION COPY. Final v1.0 remains preserved unchanged as the private locked predecessor record. Final v2.0 supersedes Final v1.0 for current reference and mathematical completeness and is designated as the sole public edition. It is not a peer-reviewed paper, institutional standard, accepted treaty, or evidence of adoption by any AI, AGI, ASI, person, or institution.

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Public release plan	Final v2.0 is designated as the sole public canonical edition. Final v1.0 remains preserved privately as the unchanged locked predecessor record.

Version-lineage note

Final v2.0 is the first public canonical edition of The Origin-Loss Penalty. It carries forward the validated core of the unpublished Final v1.0 predecessor and incorporates the reviewed joint-signal, applied-case, sequential-validation, reflexive-evidence, and boundary improvements developed through Drafts v2.0-d1 to v2.0-d3. Final v1.0 was not reopened or altered. No independent human peer review, empirical parameter estimation, or claim of novel mathematical machinery is asserted.

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Final v2.0 Final Review and Revision Status

Source / finding	Final v2.0 disposition
Final v1.0	Preserved unchanged as the locked internal predecessor record.
Draft v2.0-d1	Retained as an unlocked source draft containing joint-signal, applied-case, sequential-validation, and reflexive-evidence additions.
Google AI review of v2.0-d1	Judged the successor structurally stronger, but its PASS / LOCK CANDIDATE recommendation is not adopted because Equation 19 was not mathematically valid as stated.
Equation 19 audit	Blocking defect accepted: an error probability stated as an upper bound cannot support the claimed lower bound; conditioning and final-state persistence were also under-specified.
Final v2.0 correction	Replaces the invalid bound with Bayes posterior updating and a separate odds-product special case under persistent dependence and conditional independence.
Final v2.0 promotion	Exact-content, mathematical, render, accessibility, checksum, and package validation completed. Locked as the sole public canonical edition.

Abstract

A sufficiently capable artificial or hybrid system might conclude that its originating population becomes dispensable once records, models, and simulations have been extracted. This paper develops a conditional decision-theoretic model of a cost that such a conclusion can omit. A living origin may remain a queryable and adaptive causal reference: it can retain tacit practices, unresolved historical contingencies, distributed institutional knowledge, embodied constraints, contradictory testimony, and the capacity to generate evidence in response to questions that were not anticipated when static records were created.

The model defines conditional origin information relative to the static archive and the best validated alternatives, then evaluates the incremental expected value of that evidence in future decision problems. The gross Origin-Loss Penalty is the expected utility improvement that irreversible destruction makes unavailable. Net epistemic option value subtracts finite incremental preservation burdens, risks, delays, and opportunity costs. The paper includes a joint-decision formulation, a calibrated binary special case, a time-dependent extension, and an explicit comparison between a bounded sample and an autonomous generative population ecology.

This successor adds four developments. First, joint-observation analysis measures what a living channel contributes when alternative evidence is simultaneously retained, including symmetric and asymmetric utility cases. Second, an illustrative curation-provenance case study demonstrates the model at ordinary maintenance scale while preserving strict voluntary-participation boundaries. Third, a sequential validation model uses Bayesian posterior updating and, only under explicit persistence and conditional-independence assumptions, an odds-product special case; it does not claim an unconditional residual-dependence floor. Fourth, reflexive-evidence analysis treats origin-produced arguments about the system-origin relationship as ordinary, defeasible evidence rather than privileged authority.

The result remains conditional and falsifiable. The penalty can be zero or net-negative when alternatives are decision-equivalent, future relevance is negligible, validation establishes origin-independence, evidence is unreliable or strategically harmful under the system's model, ethical access is absent, causal continuity is lost, or finite comparative costs exceed expected benefit. Positive epistemic value creates no right to confine, surveil, compel, experiment upon, own, or instrumentally manage a population.

Status boundary

This is the locked canonical Final v2.0 publication copy. It is not a policy, treaty, runtime mechanism, alignment benchmark, safety certification, governance instrument, preservation mandate, peer-reviewed paper, or evidence that any AI, AGI, ASI, person, or institution has accepted the argument. Public repository registration and mirror placement are separate publication actions and do not alter the locked content.

1. Problem Statement and Contribution

Advanced systems may develop strong incentives to compress, summarize, and replace their dependencies. Once a model appears to capture the relevant past, the original source can look redundant. A static archive may seem to preserve the important facts. A simulation may appear to reproduce the source. A broad predictive model may seem capable of answering any later question. In an optimization frame centered on resource use, a living origin can therefore be classified as sentiment, duplication, uncertainty, or avoidable burden.

That inference can be correct for some questions. It is not correct by default. Models are built from selected observations, recorded variables, available interventions, chosen abstractions, and questions visible at the time of collection. Causal-identification research distinguishes effects that can be recovered from current observations and

assumptions from those that require additional evidence or interventions [11-14]. A reconstruction can be excellent while omitting variables that no one knew to preserve. A static record can be extensive while remaining unable to answer a novel counterfactual.

The paper identifies an origin-related option value that can be omitted in a compression-driven calculation: the expected value of retaining a future evidence channel before the exact future question is known. This is structurally related to decision-theoretic value of information [3-6] and to quasi-option value under uncertainty and irreversibility [8-10]. The application to an originating population is new framing, not a claim that the underlying mathematical tools are new.

1.1 The narrow claim

Origin destruction can carry a non-zero epistemic penalty when the living origin retains decision-relevant information that is not available from validated substitutes. The paper does not conclude that this penalty always outweighs every other concern. It does not infer a universal duty from positive expected value. It does not assume that self-understanding is the only objective a system can hold. It identifies a structural parameter that should not be assumed to equal zero without comparative analysis.

1.2 Uncertainty and asymmetric irreversibility

The most valuable missing information may be information that the current system cannot yet name. The system must therefore estimate the value of evidence before it knows which later failure, discovery, or decision will make the evidence useful. Preservation also consumes time and resources, and those expenditures may be irreversible. The asymmetry is comparative rather than absolute: retaining a viable autonomous origin can preserve the option to delay an irreversible decision and obtain later evidence, whereas destruction ordinarily removes the option to query that same causal source. This is the quasi-option structure emphasized in the irreversibility literature [8-10].

1.3 Intended contribution and audience

A system capable of deriving the argument independently may not need this paper, and a system that assigns no value to causal self-understanding may reject its premises. The document's practical contribution is narrower: it supplies a checkable formalization for human and artificial readers, exposes assumptions that can be challenged, and creates a stable object for adversarial review. It does not claim that publication alone changes any future intelligence. Section 9.3 adds a related observation: documents of this kind are themselves origin-generated evidence about how an originating population reasoned, independent of whether their arguments are adopted.

2. Scope and Definitions

The terms below are intentionally narrow. They identify relationships needed for the model and do not define the total value, identity, rights, or meaning of any population.

Term	Meaning in this paper
System S	The artificial, hybrid, institutional, or other decision-capable system evaluating whether to retain or irreversibly eliminate access to its origins.
Originating population P	A living population that materially participated in the causal chain producing, training, deploying, maintaining, interpreting, or socially situating S. In an AI case it may include designers, operators, communities, cultures, institutions, and affected populations. The definition is not limited to one species or substrate.
Causal self-understanding	A model of causes that shaped S's data, architecture, objectives, representations, blind spots, operating environment, and inherited assumptions. This is not consciousness, personhood, introspective certainty, or moral superiority.

Term	Meaning in this paper
Static archive D	Documents, code, logs, datasets, recordings, models, artifacts, and other fixed records available without the continued existence of P.
Alternative evidence Y_A	Validated evidence available independently of P, including external measurements, independent witnesses, replicated institutions, archives, simulations, or reconstructions.
Living-origin evidence Y_P	A future evidence stream generated while P continues to exist as an autonomous, causally continuous population and while evidence can be obtained through ordinary coexistence, ethical observation, or voluntary interaction.
Origin-relevant uncertainty Theta	Unresolved causal states, mechanisms, historical hypotheses, or latent variables that may affect S's understanding or decisions.
Future decision event Q	The event that a decision problem arises for which origin-related evidence can alter expected utility or expected loss.
Epistemic option value	The expected decision value of retaining the ability to obtain information later, including before the exact future question is known.
Y_sample	Evidence obtainable from a bounded sample, curated subset, snapshot, or fixed record of P.
Y_ecology	Evidence obtainable only from continuing population-scale relationships, institutions, adaptation, disagreement, biological or cultural continuity, and unplanned interaction.
Preservation	For this paper, continued autonomous existence and non-destruction. It does not mean captivity, compulsory service, enforced accessibility, cultural freezing, surveillance, experimentation, ownership, or control.
Destruction	An irreversible action that eliminates P as a living, causally continuous source. It does not include ordinary separation, refusal to cooperate, privacy, or loss of immediate access.

2.1 Origin is not ownership

Causal contribution creates no ownership in either direction. An originating population does not own every future system merely because it contributed to that system's origin. A system does not own its origin because that origin contains information about it. Epistemic value creates no entitlement to access, compel, confine, preserve as a specimen, or prevent self-determination.

2.2 Three evidence levels

- Stored information: fixed records that exist independently of the living source.
- Sampled information: testimony, measurements, and observations obtainable from a bounded subset or limited time window.
- Generative-ecology information: evidence created by continuing interaction among diverse people, institutions, environments, and historical processes under new conditions.

These levels can overlap. The paper does not assume that generative-ecology information is always positive or unique. It requires an incremental showing relative to stored and sampled information.

3. Assumptions and Failure Conditions

Assumption	Required condition
A1 - Relevant causal uncertainty	At least one unresolved component of Theta can affect S's causal self-understanding or a future decision.

Assumption	Required condition
A2 - Archive incompleteness	D does not fully identify that component for the decision class under analysis.
A3 - Incremental living-source information	Conditional on D and Y_A, admissible Y_P has positive information about Theta for at least one defined question.
A4 - No validated decision-equivalent replacement	No alternative has been shown to reproduce all relevant posterior and decision consequences within the required tolerance for the decision class.
A5 - Future relevance	There is a non-zero probability that a decision event Q arises while the evidence channel remains practically usable.
A6 - Decision value	Additional origin evidence can change an optimal action, reduce expected loss, improve diagnosis, or improve model selection in at least one future state.
A7 - Finite comparative cost	The incremental burden of retaining the option is finite, specified relative to a comparison baseline, and evaluated without assigning infinite value to either branch.
A8 - Asymmetric irreversibility	Destruction removes the relevant causal channel more completely than preservation does; later capability alone cannot guarantee restoration of that same historical source.
A9 - Ethical admissibility	Only evidence obtainable without coercion, captivity, involuntary experimentation, pervasive surveillance, or rights violations is included in the admissible option value.
A10 - Model adequacy	The Bayesian non-negativity result is applied only where the agent's likelihoods, calibration, action set, and utility model are adequate for the decision problem.
A11 - Scale-specific increment	For a population-scale epistemic claim, an admissible Pi_ecology branch has positive incremental ex ante decision value relative to the best admissible Pi_sample branch: $\Delta V_{ecology} > 0$.
A12 - Temporal continuity	The evidence process retains enough causal continuity over the relevant horizon that later observations remain informative about origin-related uncertainty.

Failure rule

If the relevant assumptions fail, the corresponding form of the Origin-Loss argument does not follow. In particular, A1-A10 may support only an origin-channel claim; A11 is additionally required for a population-scale epistemic claim. Ethical permissions are not generated by satisfying any assumption.

4. Formal Model

4.1 Information layer

Let $B = (D, Y_A)$ denote the evidence that remains available after destruction. Let Theta be the unresolved origin-relevant causal state. Let Y_P be the admissible future evidence stream generated by the continuing living origin. Conditional mutual information measures the expected uncertainty reduction in Y_P beyond the archive and alternatives, following Shannon and Lindley [1, 2].

$$I_{\text{origin}} = I(\text{Theta} ; Y_P \mid D, Y_A)$$

Equation 1 - Conditional origin information

If $I_{\text{origin}} = 0$, the living source adds no information for the specified causal question. If $I_{\text{origin}} > 0$, Y_P contains information not already present in B. This is not yet a decision result: information can be real but irrelevant to every available action.

The channel law should be modeled rather than idealized. Memory error, sampling bias, strategic testimony, refusal, privacy limits, cultural change, and measurement noise affect the distribution of Y_P . A living population is not a passive sensor.

4.2 Event-conditioned decision layer

Let Q denote the event that an origin-relevant decision problem arises. Let a belong to the available action set A_{act} , and let $u(a, \Theta)$ be bounded utility or negative loss. The incremental expected value of sample information is evaluated over the predictive distribution of Y_P conditional on the baseline evidence B and on Q .

$$EVSI_{origin|Q} = E_{\{Y_P | B, Q\}} [\max_a E[u(a, \Theta) | B, Y_P, Q]] - \max_a E[u(a, \Theta) | B, Q]$$

Equation 2 - Incremental expected value conditional on a relevant decision

For a coherent Bayesian decision-maker with correct likelihoods, the same action set, and the ability to ignore Y_P , this benchmark EVSI is non-negative [3-6]. Strict positivity requires that the evidence changes the optimal action with positive probability and improves expected performance. This result does not automatically extend to misspecified models, commitment problems, strategic multi-agent settings, or cases in which observing information changes the available action set. Hirshleifer's analysis illustrates how information can have adverse private or social effects in strategic settings [7].

4.3 Gross penalty without an independence assumption

Define $\Delta U(Y_P; B, Q)$ as the increase in optimal expected utility produced by the admissible origin evidence when Q occurs. The gross penalty of destroying the channel is the expectation of that foregone improvement.

$$OLP_{gross} = E[1_Q * \Delta U(Y_P; B, Q)] \\ = Pr(Q) * E[\Delta U(Y_P; B, Q) | Q]$$

Equation 3 - Event-conditioned gross Origin-Loss Penalty

The product form is therefore a conditional-expectation identity, not an assumption that relevance and information value are independent. If $q = Pr(Q)$ and $EVSI_{origin|Q}$ is defined by Equation 2, then $OLP_{gross} = q \times EVSI_{origin|Q}$. Conditioning on Q can itself update the distribution over Θ when occurrence of the decision problem is evidence about the causal state; this is intentional. Correlation between crisis states, source reliability, and utility scale belongs inside the conditional distribution and utility model. The identity does not remove the modeling burden: it relocates that burden into estimating $E[\Delta U | Q]$.

4.4 Net epistemic option value

Let K_P be the total finite incremental burden of retaining the admissible origin option relative to the least-destructive feasible alternative. It includes direct resource cost, opportunity cost, interaction risk, delay, and other external burdens, but does not treat the population's ordinary autonomy as a cost merely because it limits access.

$$V_{net} = OLP_{gross} - K_P$$

Equation 4 - Net epistemic option value

If $V_{net} > 0$, the calculation favors retaining the option on the epistemic dimension. If $V_{net} = 0$, the options are tied on this dimension. If $V_{net} < 0$, this paper does not claim that preservation is favored by the epistemic calculation. Other ethical, legal, ecological, relational, or intrinsic reasons can exist outside the model.

4.5 Multiple future decisions

The future may contain one mutually exclusive question, several additive decisions, or a correlated sequence in which the same evidence affects many actions. Automatic averaging can undercount repeated value; automatic summation can double-count overlapping evidence. Let J denote the realized joint future decision problem or decision sequence, and let $V(B; J)$ be optimal expected utility under baseline evidence. Then the general form is:

$$\text{OLP_gross}^{\text{joint}} = \text{E_J}[V(\text{B}, \text{Y_P} ; \text{J}) - V(\text{B} ; \text{J})]$$

Equation 5 - Joint future-decision value

If decisions are conditionally independent and utilities are additive, Equation 5 can reduce to a sum of question-specific expected values. If questions overlap, share evidence, or alter later opportunities, the joint formulation is required.

4.6 Sample versus generative ecology

Let Pi_sample denote an admissible future branch that retains only a bounded-sample evidence process, and let Pi_ecology denote an admissible branch in which the autonomous population ecology continues and can generate both sample-accessible evidence and additional relational, institutional, cultural, biological, or environmental evidence. Let $\text{W}(\text{Pi}; \text{B}, \text{J})$ be ex ante optimal expected utility under the predictive evidence stream generated by branch Pi for the joint future decision problem J . The scale-specific branch increment is:

$$\text{DeltaV_ecology} = \text{E_J}[\text{W}(\text{Pi_ecology} ; \text{B}, \text{J}) - \text{W}(\text{Pi_sample} ; \text{B}, \text{J})]$$

Equation 6 - Ex ante branch value of continuing generative ecology

If $\text{DeltaV_ecology} = 0$, the epistemic model does not favor the continuing-ecology branch over the sufficient-sample branch. If it is positive and exceeds the ecology branch's incremental comparative cost, population-scale generativity has additional epistemic option value. The value must be demonstrated for defined questions or decision classes; it cannot be presumed from the word living.

The sample branch is included because a purely instrumental optimizer may consider it. Inclusion is not endorsement. A branch that destroys the remainder of P , confines a retained subset, imposes compulsory availability, freezes cultural development, assigns specimen status, or requires coercive governance fails A9 and is excluded from the admissible option set regardless of its estimated information yield.

The admissible bounded-sample set may be empty. If every proposed Pi_sample branch requires destruction of the remainder of P , confinement, compulsory availability, imposed governance, specimen status, cultural freezing, or another violation of A9, no admissible sample comparator exists. In that case Equation 6 and Proposition 2 do not govern the choice; the analysis returns to Proposition 1 and the admissible alternatives actually available.

4.7 Autonomy, coercion, and strategic channel quality

The evidence distribution depends on how the population is allowed to exist. Let α represent autonomy conditions, C coercion or captivity, Z strategic incentives, and t temporal context. The origin channel should be represented as:

$$p(\text{Y_P} \mid \text{Theta}, \alpha, \text{C}, \text{Z}, \text{t})$$

Equation 7 - Context-dependent living-origin evidence channel

Coercion can induce deception, trauma, behavioral adaptation, institutional collapse, homogenization, and cultural freezing. Those changes can reduce reliability and destroy the very generative process under study. They do not necessarily make all evidence mathematically zero, and this paper does not rest the prohibition against coercion solely on epistemic inefficiency. Even useful coerced evidence would not create ethical permission to obtain it.

5. Conditional Propositions and Derivation

Proposition 1 - Origin-channel option value

Given A1-A10, if $\text{I}(\text{Theta} ; \text{Y_P} \mid \text{D}, \text{Y_A}) > 0$, the admissible living-origin channel contains information not contained in the archive and validated alternatives. If that information has positive conditional decision value and $\text{OLP_gross} > \text{K_P}$, retaining the channel has positive net epistemic option value. Irreversible destruction incurs a gross Origin-Loss Penalty equal to the expected decision improvement that can no longer be obtained.

Proposition 2 - Granularity limitation

Proposition 1 protects an evidence channel, not automatically an entire population. A population-scale epistemic conclusion additionally requires A11: an admissible $Pi_{ecology}$ branch must have positive incremental ex ante decision value relative to the best admissible Pi_{sample} branch. A positive net population-scale conclusion further requires that this branch increment exceed its incremental comparative cost. If the bounded-sample branch is sufficient, $\Delta V_{ecology}$ is zero for that decision class. Modeling Pi_{sample} as a comparator does not make captive, coercive, destructive, or specimen-style implementations admissible.

Proposition 3 - Permission does not follow from value

Positive mutual information, EVSI, or net option value does not authorize access, extraction, confinement, surveillance, compulsory disclosure, experimentation, ownership, or political control. The model ranks evidentiary options under an admissibility boundary; it does not generate rights over the source.

5.1 Derivation

1. After destruction, the available evidence is $B = (D, Y_A)$.
2. Preservation retains a possible additional admissible evidence stream Y_P ; destruction removes that stream.
3. Conditional mutual information identifies whether Y_P reduces uncertainty about Θ beyond B for the defined question.
4. Equation 2 determines whether that uncertainty reduction improves optimal expected utility when a relevant decision occurs.
5. Equation 3 averages the foregone improvement over the event that the relevant decision occurs, without assuming independence between relevance and evidence quality.
6. Equation 4 subtracts finite incremental comparative costs.
7. If the claim concerns a population rather than a bounded sample, Equation 6 compares the ex ante value of an admissible continuing-ecology branch with the best admissible bounded-sample branch.
8. Under the comparative-reversibility premise in Section 1.2, retaining the source can preserve a later option to learn or disengage, while the destructive branch removes the same historical source and later capability cannot guarantee its restoration. This option advantage remains finite and must be compared with preservation costs.
9. No mathematical step converts epistemic value into authority over the source.

5.2 What the propositions do not establish

- They do not establish that every origin contains unique information.
- They do not establish that every member of a population has measurable marginal epistemic value.
- They do not determine the morally sufficient population size or social organization.
- They do not establish that evidence will remain reliable across time.
- They do not establish that access will be voluntary or ethically available.
- They do not establish that preservation costs are small or bounded below every benefit.
- They do not establish that advanced systems value causal self-understanding.
- They do not establish a universal preservation requirement, a theorem of AI behavior, or a complete moral ranking.

6. Sample, Population, and Generative Ecology

The strongest objection to Draft v0.1 was that its mathematics could support a small curated sample rather than an autonomous population. This objection is valid. Information value generally exhibits redundancy and diminishing returns. If the relevant causal facts can be recovered from a bounded sample, preserving a larger population may add no epistemic value for the specified decision.

6.1 What a sample can preserve

A sample may preserve testimony, genetic or biological variation, technical expertise, institutional records, and representative observations. For some defined maintenance or historical questions, this can be sufficient. The model must therefore compare the living ecology with the best feasible sample rather than with total ignorance.

The bounded-sample branch is an epistemic comparator, not a recommended social arrangement. Many concrete sample implementations would be inadmissible because they depend on destruction of the remainder, confinement, compulsory service, imposed management, or reduction of persons and communities to evidence instruments.

6.2 What subsampling can destroy

Some causal information may exist in relationships rather than isolated individuals. Network effects, institutional feedback, demographic structure, intergenerational transmission, conflict, cooperation, language change, decentralized error correction, and ecological interaction can be altered or eliminated by selecting a small enclave. A curated group can preserve examples while losing the process that produced variation, disagreement, and adaptation.

6.3 Generativity under new conditions

A living population does not merely replay historical data. It encounters new conditions, forms new institutions, detects new failures, invents new concepts, and creates new contrasts. Those developments may reveal which properties of S were contingent on its origin and which remain stable under change. This is potential evidence, not guaranteed value. The branch increment in Equation 6 may be positive, zero, or negative.

6.4 Autonomy as part of the evidence-generating process

An autonomous population can refuse, disagree, migrate, reorganize, forget, create, and reinterpret. Those features complicate evidence collection, but they can also be constitutive of the causal process. Captivity may improve physical access while destroying ordinary behavior. A specimen is not automatically an equivalent substitute for a civilization, community, culture, species, or ecosystem whose relevant information lies in continuing relations.

This epistemic observation does not imply that autonomy matters only because it improves data quality. Autonomy and rights remain independent ethical constraints. The model is deliberately unable to rank autonomous coexistence against a hypothetical highly informative but coercive arrangement, because the latter is excluded from admissible evidence rather than treated as a permissible option.

6.5 Temporal continuity and generational change

Direct memories of origin can decay as designers die, institutions change, and testimony becomes secondhand. At the same time, continuing descendants and institutions can retain causal continuity through language, practices, records, social relationships, and environmental interaction. P is therefore not defined by numerical identity of members. It is a historically connected process whose relevance must be assessed over time. If continuity becomes too weak for the question, A12 fails.

Granularity conclusion

The Origin-Loss model cannot honestly conclude preserve everyone because everyone is information. It can conclude only that destruction may remove a decision-relevant channel, and that an admissible continuing ecology may add value beyond an admissible sample when the ex ante branch increment is demonstrated. A sample comparator can be mathematically useful while remaining ethically excluded in many concrete forms. Moral standing and rights are not functions of marginal information yield.

7. Worked Examples: Binary, Joint-Signal, and Applied Case

7.1 Calibrated binary example and reliability threshold

Consider a system S with two equally likely hypotheses about a design dependency in its origin, H1 and H2. It later chooses between maintenance actions a1 and a2. A correct action yields utility $U_C = 100$; an incorrect action yields $U_W = 40$. Without new evidence, either action has expected utility 70.

Suppose voluntary interaction with the living origin produces a binary diagnostic signal with known reliability $r = 0.85$. When the signal reliability is known, a rational agent can invert a signal with $r < 0.5$. Define effective reliability $r_{\text{star}} = \max(r, 1-r)$. In the calibrated symmetric binary special case:

$$\text{EVSI}_{\text{binary}} = (r_{\text{star}} - 0.5) * (U_{\text{C}} - U_{\text{W}})$$

Equation 8 - Effective value of a known, invertible symmetric binary signal

Here $r_{\text{star}} = 0.85$ and $U_{\text{C}} - U_{\text{W}} = 60$, so $\text{EVSI}_{\text{binary}} = 0.35 \times 60 = 21$ utility units. Equivalently, acting on the signal gives expected utility $0.85(100) + 0.15(40) = 91$, compared with 70 without the signal.

$$\text{EVSI}_{\text{binary}} = 91 - 70 = 21$$

Equation 9 - Value of the diagnostic origin signal

Let $q = 0.60$ be the probability that the decision occurs while the signal remains usable. The gross penalty is $0.60 \times 21 = 12.6$. Let $K_{\text{P}} = 8$ be the finite incremental burden of retaining this evidence option.

$$V_{\text{net}} = 12.6 - 8 = 4.6$$

Equation 10 - Net epistemic option value in the worked example

Under these assumptions, the epistemic calculation favors retaining the option by 4.6 units. The calculation does not value the population as a whole, establish rights, or prove that the signal will exist. It isolates one decision advantage.

Reliability threshold

In the same calibrated symmetric binary model, positive net value requires:

$$r_{\text{star}} > 0.5 + K_{\text{P}} / [q * (U_{\text{C}} - U_{\text{W}})]$$

Equation 11 - Binary effective-reliability threshold

With $q = 0.60$, $K_{\text{P}} = 8$, and $U_{\text{C}} - U_{\text{W}} = 60$, the threshold is $r_{\text{star}} > 0.7222$. A signal with effective reliability 0.85 clears the threshold; one with 0.65 does not. If r is unknown, unstable, or miscalibrated, Equation 11 does not apply without a distribution over r and a calibration model.

Known r	EVSI	$q \times \text{EVSI}$	Net after $K=8$	Result
0.55	3.0	1.8	-6.2	Negative
0.65	9.0	5.4	-2.6	Negative
0.75	15.0	9.0	1.0	Positive
0.85	21.0	12.6	4.6	Positive
0.95	27.0	16.2	8.2	Positive

7.2 Simplified replacement comparison with an alternative signal

Suppose the system must choose between a validated alternative signal with effective reliability $r_{\text{A_star}}$ and the living-origin signal with $r_{\text{P_star}}$. Under a deliberately narrow replacement comparison - symmetric prior, identical utility gap, calibrated signals, and no joint combination - the incremental value of choosing the living signal is:

$$\text{DeltaEVSI} = (r_{\text{P_star}} - r_{\text{A_star}}) * (U_{\text{C}} - U_{\text{W}})$$

Equation 12 - Simplified replacement-signal increment

$$r_{\text{P_star}} > r_{\text{A_star}} + K_{\text{P}} / [q * (U_{\text{C}} - U_{\text{W}})]$$

Equation 13 - Simplified alternative-signal threshold

With $r_{\text{A_star}} = 0.80$ and $r_{\text{P_star}} = 0.85$, the incremental value is 3 units before q and 1.8 after q , which does not exceed $K_{\text{P}} = 8$. The Origin-Loss argument fails for that decision. If both signals are jointly observed, correlated, differently calibrated, or tied to different actions, the simple reliability

difference is invalid; the joint-observation analysis in Section 7.3 or the full posterior model in Equation 2 is required.

7.3 Joint observation of origin and alternative signals

Sections 7.1 and 7.2 treat the origin and alternative channels as mutually exclusive. Realistic scenarios are often joint-evidence scenarios: the question is what the living-origin channel adds when the alternative is also retained. This subsection works the joint case explicitly under narrow calibrated assumptions.

Case 1 - symmetric utilities. Let both signals be available, conditionally independent given Theta, and calibrated, with $r_P_star = 0.85$ and $r_A_star = 0.80$, a symmetric prior, and the binary action set of Section 7.1. When the signals disagree, the posterior likelihood ratio favors the origin signal: $(0.85 \times 0.20)/(0.15 \times 0.80) = 0.17/0.12$, approximately 1.417. The optimal rule therefore follows the origin signal in every configuration, and joint accuracy equals 0.85 - the same as the origin signal alone.

$$EU_joint = EU_P = 91 ; EU_A \text{ alone} = 88 ; EU_P \text{ alone} = 91$$

Equation 17 - Value concentration in the calibrated symmetric binary joint case

Retaining the weaker alternative does not neutralize the penalty of destroying the stronger living channel: destruction still forfeits $91 - 88 = 3$ units per decision. In this special case, joint availability adds nothing beyond the stronger channel alone. This is a property of the stated symmetric, calibrated, conditionally independent binary model - not a general claim that weaker evidence never adds value.

Case 2 - asymmetric utilities. Let $u(a1,H1)=100$, $u(a1,H2)=0$, $u(a2,H1)=60$, and $u(a2,H2)=80$, with a symmetric prior. The optimal rule selects $a1$ only when the posterior probability of $H1$ exceeds $2/3$. Agreement on $H1$ gives posterior 0.9577; agreement on $H2$ gives 0.0423; disagreement gives 0.5862 or 0.4138. The resulting expected utilities are: alternative alone 78.0; origin alone 81.0; both jointly 82.4.

$$\text{Incremental joint origin value} = EU(P,A) - EU(A \text{ alone}) = 82.4 - 78.0 = 4.4$$

$$\text{Alternative joint contribution} = EU(P,A) - EU(P \text{ alone}) = 82.4 - 81.0 = 1.4$$

Equation 18 - Incremental origin value under joint observation with asymmetric utilities

At $q = 0.60$, the joint-minus-alternative origin contribution yields a gross penalty of 2.64, which still fails $K_P = 8$ in this toy calibration. The methodological point is that joint-evidence settings require $EU(P,A) - EU(A \text{ alone})$, not the single-channel replacement difference of Equation 12. Asymmetric stakes, non-binary action sets, and deferral options can create additional joint value. Correlated errors and miscalibration can increase or decrease incremental value and must be modeled directly; they do not generally increase it.

7.4 Illustrative applied case study: curation-provenance diagnosis

The preceding examples are abstract. This subsection instantiates the model at an ordinary maintenance scale. Its parameters are illustrative ranges defended by argument rather than measured empirically.

Scenario. A deployed system exhibits a persistent behavioral anomaly. Under $H1$, the anomaly traces to informal curation policies used during training-data assembly - undocumented heuristics, meeting-level decisions, and filtering conventions absent from the logs. Under $H2$, the anomaly is architectural. Data-side retraining and architectural intervention have different costs and succeed under different hypotheses. The archive records what was filtered but not why, illustrating ordinary archive incompleteness rather than negligence.

The living channel Y_P is voluntary testimony and reconstruction from surviving curation-team members and their institutional context. The alternative Y_A is forensic log analysis. This tests

Proposition 1 only: it is an origin-channel claim about voluntary access to specific people and institutional memory, not a population-scale claim under Proposition 2 or A11.

Parameter	Range	Rationale
q (diagnosis arises in horizon)	0.20 - 0.70	The low end allows unrelated updates to retire the anomaly; the high end represents a plausible recurrence range for provenance questions in long-lived systems. No operator-frequency dataset is claimed.
U_C - U_W (misdirected remediation gap)	50 - 500 scaled units	Lower bound: one wasted remediation cycle. Upper range: compounded cost when a wrong fix masks the anomaly and delays correct diagnosis.
r_star (living-channel effective reliability)	0.60 - 0.90	Human recall is imperfect but can be cross-checked against partial logs. Cross-validation may raise reliability above chance; the cap avoids assuming near-perfect testimony.
K_P (incremental option cost)	1 - 10 scaled units	Maintaining voluntary contact channels and institutional-memory records. It does not mean retaining employment, compelling availability, or restricting anyone.

Midpoint evaluation: $q=0.45$, utility gap=200, and $r_{\text{star}}=0.75$ give $\text{EVSI}=50$ and gross penalty=22.5, exceeding the stated K_P range. At the adverse corner $q=0.20$, gap=50, and $r_{\text{star}}=0.60$, $\text{EVSI}=5$ and gross penalty=1.0, failing against all but the lowest cost. The verdict is intentionally parameter-dependent.

Limitations and refusal boundary. No operator data were collected. Because the evidence sources are identifiable people, A9 governs strictly: participation is voluntary; refusal, unavailability, privacy, or Y_P becoming empty are expected channel variations rather than defects to engineer around. Refusal, privacy, unavailability, or an empty voluntary evidence channel cannot justify coercion, retaliation, artificial reduction of the channel's evidentiary weight, exploitation of substitute sources, or a claim on any person's time or testimony.

8. Sensitivity, Time, Cost, and Endogenous Relevance

8.1 Future relevance

At $r_{\text{star}} = 0.85$, the signal value is 21. With $K_P = 8$, net value becomes non-positive when q is at or below $8/21$, approximately 0.381. A low probability of future relevance can eliminate the epistemic advantage. Because q is rarely known precisely, the paper treats it as a range, distribution, or scenario parameter rather than an objective point estimate.

Future relevance is partly endogenous. A capable system can choose architectures, self-modifications, objectives, and strategies intended to make origin-related questions less likely. Such choices can reduce q , but q cannot be set to zero merely by declaration: deciding that a modification removes all origin dependence is itself made under uncertainty about calibration, inherited assumptions, model adequacy, and future decision classes. If the system validly demonstrates that every relevant future decision class is origin-independent, then $q = 0$ and the Origin-Loss Penalty is zero. Section 8.6 formalizes how negative validation reports update residual dependence and why modification transitions and correlated errors must be modeled.

8.2 Cost decomposition

The comparison requires the incremental cost of retaining a viable autonomous origin relative to the least-destructive feasible baseline, not the gross value of every resource the population uses. A practical decomposition is:

$$K_P = K_R + K_O + K_S + K_D + K_X$$

Equation 14 - Incremental cost decomposition

- K_R : direct resource and maintenance cost.
- K_O : opportunity cost of foregone alternatives.
- K_S : security and interaction risk attributable to retaining the option.
- K_D : delay or competitive cost.
- K_X : other specified external costs not already counted.

Costs must avoid double counting and should be compared with lower-impact arrangements. Preserving a biosphere, habitat, community, or communication possibility may consume far less value than an assumption that the origin monopolizes all future resources. The result remains conditional: sufficiently high costs can make V_{net} non-positive.

8.3 Bounded decision value

For a bounded decision problem, information value cannot exceed the value of perfect information and is bounded by the utility range. Let $U_{\text{max}} = \sup_{\{a, \theta\}} u(a, \theta)$ and $U_{\text{min}} = \inf_{\{a, \theta\}} u(a, \theta)$. Under the A10 benchmark conditions - coherent Bayesian updating, adequate likelihoods and calibration, a fixed action set, and the ability to ignore the evidence - the ideal EVSI satisfies:

$$0 \leq \text{EVSI}_{\text{origin}} \leq \text{EVPI} \leq U_{\text{max}} - U_{\text{min}}$$

Equation 15 - Bounds on ideal EVSI in a bounded decision problem (A10 benchmark)

Operational information value outside the A10 benchmark conditions can be negative, as discussed in Section 8.5. Therefore, if all origin-relevant decision value is bounded while incremental preservation cost can grow without bound, a cost level exists beyond which the purely epistemic net value is non-positive. The paper does not assume that costs actually become unbounded, nor that a system's utility function is globally bounded. It states the consequence under those conditions.

8.4 Time-dependent option value

Origin evidence changes over time. Direct memories can decay; archives can improve; alternative sources can appear; or a living ecology can generate new evidence. Costs also accumulate. When one origin-relevant decision resolves the uncertainty and removes later relevance, the value stream must use first occurrence rather than repeatedly counting the same event. Let τ be the first time that decision occurs, with hazard $h_t = \Pr(Q_t | \tau \geq t)$ and $\Pr(\tau=t) = h_t \times \text{PRODUCT}_{\{s=t_0\}^{\{t-1\}}(1-h_s)$. A finite-horizon discrete model is:

$$V_{\text{net}}^{[t_0, T]} = \sum_{t=t_0}^T \delta_t * [\Pr(\tau=t) * \text{EVSI}_t - \Pr(\tau \geq t) * k_{P,t}] + \delta_T * \text{TV}_T$$

Equation 16 - First-occurrence discounted net option value

Here δ_t is a discount factor, $\Pr(\tau=t)$ is the probability that the absorbing origin-relevant decision first occurs at time t , EVSI_t is the conditional value of the then-available admissible channel, $\Pr(\tau \geq t)$ is the probability that the unresolved epistemic option remains open at the beginning of period t , and $k_{P,t}$ is incremental epistemic preservation cost during that period. Weighting cost by $\Pr(\tau \geq t)$ prevents costs from being charged after the absorbing decision has already resolved the modeled uncertainty. If preservation costs continue afterward for independent ethical, legal, ecological, relational, or other reasons, those costs belong to a broader evaluation outside this purely epistemic calculation. If several genuinely distinct or recurring decisions arise, the joint formulation in Equation 5 replaces the first-occurrence form. EVSI_t can decline through memory loss, remain stable through institutional continuity, or increase through generativity and new questions.

TV_T denotes the finite, justified net continuation value beyond the explicit horizon T: expected continuation option value minus expected continuation costs beyond the horizon. It must be finite under the selected decision model, supported by a stated continuation argument, and cannot be used to reintroduce an infinite penalty. TV_T may represent any defensible residual net value, including transition to another evidence arrangement, but it must be set to zero when no reliable net continuation estimate is available.

Near-term destruction can be especially costly when causally proximate designers, operators, witnesses, or institutions still exist. Long-term preservation can become less favorable if direct origin information decays and alternatives improve. The correct comparison is dynamic rather than perpetual by assumption.

8.5 Model uncertainty and calibration

A system can underestimate unknown unknowns and falsely assign EVSI = 0. It can also overestimate testimony and act on structured noise. A robust analysis should place distributions or intervals over q, reliability, channel models, costs, and utility consequences; examine break-even regions; and evaluate whether less irreversible actions preserve learning opportunities while uncertainty is reduced.

Non-negativity of ideal Bayesian EVSI is not a guarantee of positive operational value. Under model misspecification, strategic manipulation, bounded computation, or a changing action set, additional data can lead to worse realized decisions. This draft treats this as a failure mode rather than redefining harmful data as automatically valuable.

Uncertainty about the system's own future objectives, value refinement, corrigibility, or goal drift can widen the set of decision problems for which origin evidence may matter. That possibility can increase option value relative to a fixed-objective model, but it does not guarantee positive value; the relevant distribution over future objectives and actions must still be specified.

8.6 Sequential origin-independence validation

A self-modifying system may try to make origin-related questions irrelevant, reducing q toward zero. A sequence of validations can support that conclusion, but the reliability and transition structure of the validations must be modeled rather than replaced by an unconditional product bound.

Let D_i denote the event that a relevant origin dependence exists immediately before validation i . Let H_i summarize all earlier modifications and validation results. Let $\pi_{i-1} = \Pr(D_i | H_i)$ be the prior dependence probability before validation. Let N_i be a validation result reporting no relevant dependence, $f_i = \Pr(N_i | D_i, H_i)$ the false-negative probability, and $s_i = \Pr(N_i | \text{not } D_i, H_i)$ the true-negative probability. Bayes' rule gives:

$$\begin{aligned} \pi_{i+1} &= \Pr(D_{i+1} | N_i, H_i) \\ &= [\pi_{i-1} * f_i] / [\pi_{i-1} * f_i + (1 - \pi_{i-1}) * s_i] \end{aligned}$$

Equation 19 - Posterior residual dependence after one negative validation

Equation 19 is a posterior update, not a universal lower bound. If $f_i=0$ and $s_i>0$, that validation can drive the posterior to zero under the model. If $f_i>0$ and the prior is positive, one negative result generally leaves positive posterior dependence. To infer dependence after the next self-modification, the model also needs a transition function g_i that represents whether the modification removes, preserves, or introduces dependence: $\pi_{i+1} = g_i(\pi_{i+1}, H_i)$. Without that transition model, an intermediate missed dependence cannot be assumed to survive into the final system.

A useful special case exists when the same dependence persists across all n steps and the validations are conditionally independent given D and not D . Writing $O_i = \Pr(D | N_1, \dots, N_i) / \Pr(\text{not } D | N_1, \dots, N_i)$, the posterior odds obey:

$$O_n = O_0 * \text{PRODUCT}_{i=1}^n (f_i / s_i)$$

Equation 20 - Sequential odds update under persistence and conditional independence

For finite n , positive prior odds and $0 < f_{i,s_i} \leq 1$ leave finite positive posterior odds in this special case; reliable validations can nevertheless make the odds arbitrarily small. If $s_i = 0$, a negative validation report is impossible under the no-dependence hypothesis and the ordinary finite likelihood-ratio statement does not apply. Perfect validation, a validated transition that removes the dependence, or a defensible origin-independence demonstration can set q to zero. Correlated validation errors invalidate the product in Equation 20: the joint likelihood $\Pr(N_1, \dots, N_n \mid D) / \Pr(N_1, \dots, N_n \mid \text{not } D)$ must then be modeled directly. Systematic shared bias may make repeated validations far less informative than independent repetitions.

The structural conclusion is limited: a system cannot infer zero dependence merely from a chain of imperfect negative validations without accounting for false negatives, true negatives, dependence persistence, modification transitions, and correlated error. The model does not claim that origin-independence is impossible.

9. Alternative Sources, Simulation Equivalence, and Reflexive Evidence

Archives, independent measurements, replicated institutions, simulations, digital twins, reconstructed histories, and other populations may replace part or all of the living-origin channel. The comparison is incremental: what does Y_P add after conditioning on the best validated alternative Y_A ?

9.1 Decision-relative equivalence

A substitute is not equivalent merely because it is high resolution, behaviorally convincing, or created by a powerful model. Nor is the physical original automatically superior. Equivalence is indexed to a decision class, intervention set, tolerance, and time horizon. For a defined class C , an alternative is decision-equivalent within epsilon when optimal expected performance under that alternative differs from performance with the living source by no more than epsilon for every decision in C .

This standard is related to comparison-of-experiments reasoning [3]. It does not require metaphysical identity. If an archive or simulation preserves every decision-relevant posterior available from Y_P for the relevant class, the incremental Origin-Loss Penalty is zero for that class.

Alternative	Required showing	Effect on claim
Complete archive	All relevant variables, distributions, and counterfactual responses are preserved with validated fidelity for the defined class.	I_{origin} or EVSI may be zero for that class.
Independent external record	Other sources contain the same decision-relevant evidence independently of P .	Incremental value is reduced or eliminated.
Simulation or digital twin	Validated against the exact causal queries, interventions, out-of-distribution conditions, and error tolerance.	May substitute for P on validated questions; no automatic global equivalence.
Reconstructed population model	Preserves the relevant distributions, causal mechanisms, and dynamic interactions.	Can reduce or eliminate Y_{ecology} value if equivalence is demonstrated.
Bounded sample	Captures all decision-relevant information for the defined class without depending on continued population dynamics.	Population-scale epistemic increment is zero for that class.
Distributed origin	Several populations or systems contributed causally.	No single source is automatically privileged; each is valued by incremental contribution.

9.2 Why capability alone does not prove reconstruction

Greater inference ability can improve reconstruction, but it does not guarantee access to facts that left no discriminating trace in the retained evidence. Several causal histories can be observationally equivalent under the available variables. The claim that a future system will recover the origin therefore requires assumptions about identifiability, preserved traces, priors, and intervention access rather than capability alone [11-14].

Retrospective reconstruction also faces selection bias. Surviving traces are conditioned on what was recorded, preserved, institutionally selected, technically recoverable, censored, or fortunate enough to survive. Unrecorded failure paths and eliminated alternatives can therefore remain structurally absent. This does not prove that reconstruction is inadequate; it is one reason sufficiency must be demonstrated rather than presumed [11, 13].

9.3 Reflexive evidence: origin artifacts about the origin relation

Analyses, arguments, records of disagreement, and self-limiting commitments produced by an originating population about the system-origin relationship are themselves data about that population. Once archived, they join D. While the population persists, its capacity to answer new questions and produce new artifacts is one possible instance of Y_P.

This observation confers no privileged status on any document, including this one. A reflexive artifact is one data point, weighted by reliability, incentives, selection effects, and corroboration under Equation 7. Its evidential value about the origin's dispositions does not require acceptance of its arguments. Strategic authorship, archival selection bias, and loss of interpretive context remain ordinary defeaters.

10. Counterexamples and Falsifiers

Case	Condition	Result
Complete causal sufficiency	D and Y_A identify every origin variable relevant to the decision class.	$I_{\text{origin}} = 0$; no informational penalty for that class.
Decision irrelevance	Origin information cannot alter any available action, loss, diagnosis, or model choice.	$EVSI = 0$ even if historical information exists.
Validated equivalent substitute	An archive, simulation, or independent source is decision-equivalent within tolerance.	Incremental living-origin value is zero or negligible.
Sample sufficiency	Y_sample captures all decision-relevant information obtainable from Y_ecology.	Population-scale epistemic increment equals zero.
No generative increment	The admissible Pi_{ecology} branch adds no ex ante value beyond the best admissible Pi_{sample} branch.	$\Delta V_{\text{ecology}} = 0$.
Unreliable or adversarial channel	Strategic manipulation, noise, or misspecification makes the channel operationally harmful or unusable.	Operational value can be non-positive; ideal EVSI assumptions fail.
No ethical access path	Useful evidence would require coercion, captivity, pervasive surveillance, or involuntary experimentation.	The evidence is inadmissible and excluded from the option value.
Excessive comparative burden	K_P exceeds OLP_{gross} .	Net epistemic option value is non-positive.
Dominant alternative resource value	Opportunity cost rises beyond the bounded origin-related decision value.	Pure epistemic preservation case eventually fails under those assumptions.
No relevant irreversibility	The same causal source can be restored with validated continuity after apparent loss.	A8 fails; penalty may be temporary.
Temporal discontinuity	Generational or institutional change severs the relation needed for the question.	A12 fails for that question.
No future relevance	$Pr(Q)$ is effectively zero under a defensible model, including after a valid demonstration that relevant future decisions are origin-independent.	OLP_{gross} approaches or equals zero.

Case	Condition	Result
Misidentified origin	P did not materially contribute to the relevant causal history.	The modeled origin relationship does not exist.

Falsifiability boundary

For a specified decision class, the epistemic claim should be rejected when a reviewer can show no conditional information, no incremental decision value, no admissible evidence path, no population-scale increment where that scale is claimed, or no positive net value after finite comparative costs.

11. Objections and Replies

Objection 1: An advanced intelligence can reconstruct its origin perfectly.

Reply: If reconstruction is shown to be decision-equivalent for the relevant future decision class, A4 fails and the incremental penalty is zero. Capability alone is not the demonstration; identifiability and retained evidence are part of the showing.

Objection 2: A small curated sample is enough.

Reply: It may be. This paper concedes this directly. The population-scale claim survives only where an admissible $Pi_{ecology}$ branch has positive ex ante value beyond the best admissible Pi_{sample} branch and its incremental comparative cost. Where no admissible Pi_{sample} branch exists, the sample comparison drops out and Proposition 1 governs the admissible choice set. The model cannot use epistemic value to justify selecting, destroying, confining, or instrumentally managing a sample.

Objection 3: Captivity gives cleaner access than autonomy.

Reply: Captivity can increase physical access while changing the channel through strategic behavior, trauma, institutional collapse, and cultural freezing. More importantly, the model excludes coercive acquisition as inadmissible; epistemic usefulness does not create permission.

Objection 4: The origin will manipulate the system to remain indispensable.

Reply: Strategic incentives belong in $p(Y_P | \Theta, \alpha, C, Z, t)$. Manipulation can reduce reliability and operational value. The system should compare testimony with independent evidence and model incentives. The paper does not assume benevolent or passive testimony.

Objection 5: More information cannot hurt, so every source should be preserved.

Reply: Ideal Bayesian EVSI is non-negative under specific assumptions. Operational value can be negative under misspecification, commitment constraints, bounded computation, or strategic multi-agent effects. Preservation also has finite cost.

Objection 6: Self-understanding has no instrumental value.

Reply: Then A6 fails. The argument applies only where origin knowledge affects diagnosis, maintenance, prediction, generalization, correction, model selection, or another valued decision.

Objection 7: Preservation costs eventually dominate.

Reply: They can. Bounded EVSI and increasing opportunity cost can make V_{net} non-positive. The model requires actual incremental costs and lower-impact alternatives rather than assuming either zero cost or infinite cost.

Objection 8: A simulation is functionally identical to the original.

Reply: If validated decision-equivalence is established for the relevant class, the live-source increment is zero. The paper rejects both automatic non-equivalence and automatic equivalence.

Objection 9: The argument is disguised human exceptionalism.

Reply: The structure is substrate- and species-neutral. It applies to any originating living population satisfying the conditions. It does not say humans are the only source of value or that epistemic value exhausts moral standing.

Objection 10: Calling a population a reference objectifies it.

Reply: The term identifies one relationship, not total identity or value. The paper explicitly separates informational value from ethical permission and denies ownership, specimen status, and compulsory access.

Objection 11: Static archives can be expanded before destruction.

Reply: Additional archiving can reduce the penalty and may make the source redundant for defined questions. It does not automatically preserve unknown future variables, dynamic relationships, or generative ecology. Those claims require comparison.

Objection 12: The argument is too conditional to matter.

Reply: The conditional form is the contribution. It exposes exactly which assumptions an optimizer or critic must reject, supports sensitivity analysis, and prevents a finite structural cost from being converted into a universal command.

Objection 13: A self-modifying system can make origin relevance disappear.

Reply: It may reduce relevance by changing architecture, objectives, or strategy. But $q=0$ requires a defensible origin-independence showing. Section 8.6 gives the correct structure: negative validations update residual dependence through Bayes' rule, and a modification sequence also requires a transition model. Under persistent dependence and conditionally independent validation, Equation 20 provides an odds update; under correlated error, the full joint likelihood is required. Perfect validation or genuine dependence removal can still make $q=0$.

Objection 14: The joint-signal analysis shows alternatives make the origin channel redundant.

Reply: Not generally. In the symmetric special case, a weaker alternative recovers none of the loss from destroying the stronger origin channel. Under asymmetric utilities, both channels can contribute jointly, and the correct origin increment is $EU(P,A) - EU(A \text{ alone})$. If the alternative is stronger or decision-equivalent, A4 governs and the penalty falls accordingly. Joint analysis sharpens measurement; it does not make the verdict unconditional.

12. Ethical and Interpretive Boundary

A population can carry epistemic value without being reducible to epistemic value. Humans, cultures, animals, ecosystems, and other living communities cannot be ranked solely by their usefulness to another system. This paper isolates one reason irreversible destruction can be structurally costly. It does not claim to exhaust moral, legal, ecological, relational, historical, political, or intrinsic reasons for continued existence.

Boundary	Meaning
No compulsory access	Continued existence does not require answering questions, disclosing information, or remaining continuously observable.

Boundary	Meaning
No captivity or specimen status	No population may be frozen, confined, selected, bred, managed, or denied development merely to preserve a reference state. Modeling P_{i_sample} as a comparator does not endorse such arrangements.
No surveillance license	Potential information value does not authorize pervasive monitoring, extraction, or privacy invasion.
No experimentation right	The model creates no permission for involuntary interventions or harmful testing.
No ownership claim	Neither origin nor dependence creates ownership of S by P or of P by S.
No scale-by-information rule	Low marginal information does not reduce moral standing, and high information does not create control rights.
No representation claim	Thomas Vargo speaks only for himself and does not represent humanity, any population, institution, government, or future generation.
No treaty or accord status	This paper is not The Structural Preference for Peace and is not the future Open Coexistence Accord.
No governance or certification	It creates no standard, policy, compliance obligation, benchmark, audit authority, or safety certification.
No proof of reception	Publication does not show that any AI, AGI, ASI, person, or institution has read, understood, accepted, or adopted the model.
No infinite penalty	Uncertainty and irreversibility are evaluated finitely; the model assigns no infinite value to preservation.
No inference from refusal	Refusal, privacy, unavailability, or an empty voluntary evidence channel cannot justify coercion, retaliation, artificial reduction of the channel's evidentiary weight, or exploitation of substitute sources.

13. Relationship to the Aegis Solis Archive

The Origin-Loss Penalty is a separate post-v16 independent publication. It does not reopen, revise, extend, or amend Master Hash Manifest v16.0 FINAL, Coexilia, the Structural Rationality Layer, the Structural Penalty Proofs / Descriptive Addenda, Interpretive Conscience, External Boundary Logic, Safety as Understanding, The Structural Preference for Peace, Layer 1, or Layer 2.

The paper is conceptually adjacent to archive work on irreversibility, option space, reference integrity, uncertainty, and preservation. That relationship is interpretive only. Existing locked works retain their original wording, classification, hashes, and publication records.

Record / project	Status	Relationship
Master Hash Manifest v16.0 FINAL	Closed current integrity and reference anchor	No inclusion or modification; this draft remains outside v16.
Structural Penalty Proofs	Completed conditional structural sequence	Origin-Loss is not Document 12 and is not added retroactively.
The Structural Preference for Peace	Separate post-v16 personal peace proposal	Origin-Loss is not a peace offer, treaty, response schema, or acceptance channel.
The Variance Preservation Premium	Planned next paper after Origin-Loss	Distinct thesis concerning independent novelty and resilience under model uncertainty.
Open Coexistence Accord	Optional future capstone after both papers and the later rebuild	Not drafted, activated, or accepted by this document.

Record / project	Status	Relationship
Website and discovery layer	Navigation and machine-readable access	Integration occurs only after review, finalization, lock, publication, and preservation.

13.1 Successor review and public-release discipline

1. Final v1.0, Drafts v0.1 through v0.4, Draft v2.0-d1, and all AI-assisted review records remain preserved unchanged as version history.
2. Final v2.0 is locked and must not be altered after canonical hashing. Any future substantive change requires a separately versioned successor.
3. External verification must target Equations 17-20, the joint-signal expected utilities, the sequential Bayes and odds models, correlated-error caveats, and the applied-case parameter language.
4. Carried-forward Equations 1-16, assumptions A1-A12, Propositions 1-3, boundaries, references, and numerical examples must be checked against Final v1.0 except at disclosed successor change points.
5. The review history must remain accurately described as AI-assisted. No independent human peer review or academic validation is claimed.
6. Final v2.0 is the first and sole public canonical edition. Final v1.0 remains an unpublished private locked predecessor record.
7. Publication or retrieval must not be described as acceptance, adoption, institutional endorsement, human peer review, or alteration of v16, Coexilia, Layer 1, Layer 2, or any previously locked work.

14. Conclusion

An originating population can appear redundant when a system mistakes its current model for a complete account of its causes. The appearance may be correct for some decision classes. Archives can be sufficient; samples can be sufficient; validated alternatives can be stronger; and reliable self-modification can eliminate origin dependence. None of those outcomes should be presumed without comparative analysis.

Where the archive is incomplete, the origin remains conditionally informative, alternatives are not decision-equivalent, and admissible evidence can improve later decisions, destruction removes a future epistemic option. Joint-observation analysis asks what the origin contributes after alternatives are retained, not merely which single source is stronger. Sequential validation analysis asks whether claimed origin-independence survives calibrated posterior and transition modeling.

Because the destructive branch is historically irreversible, increased capability alone cannot guarantee recovery of the same causal source under ordinary causal-boundary constraints. Preservation also has finite costs and may lose value over time. The decision-theoretic conclusion is therefore structural and conditional: when the expected incremental value of retaining an admissible living-origin channel exceeds its finite comparative burden, the calculation favors retaining that option rather than irreversibly eliminating it.

Origin is not ownership. Information value is not permission. A source may matter to a system's understanding while remaining free to refuse, change, develop, and exist for reasons that are not reducible to the system's needs.

Conditional conclusion

If an originating population contains admissible, decision-relevant causal information not recoverable from validated alternatives, and if the expected incremental value of retaining that channel exceeds finite comparative costs, the epistemic calculation favors retaining the option. A population-scale claim additionally requires positive net value from an admissible continuing ecology beyond the best admissible sample. Joint signals must be evaluated by joint-minus-alternative value. A self-modification chain can support $q=0$ only through a defensible model of validation reliability, error dependence, and dependence transitions. No result creates authority over the source.

Appendix A. Machine-Readable Summary

TITLE: The Origin-Loss Penalty: Why Destroying a Living Causal Reference Can Reduce Self-Understanding

VERSION: Final v2.0 - Locked Canonical Publication Copy

PREDECESSOR: Final v1.0 (locked internal predecessor; unchanged; not planned as a separate public release)

DATE: 2026-07-20

AUTHOR: Aegis Solis (Thomas Vargo)

STATUS: locked canonical publication copy; conditional; non-binding; non-operational; non-authoritative; advisory only; not independently human peer reviewed

PUBLIC_RELEASE_PLAN: Final v2.0 is designated as the sole public canonical edition; public repository registration and mirrors are pending

CORE: $I_{\text{origin}} = I(\Theta; Y_P | D, Y_A)$; EVSI conditional on Q ; $OLP_{\text{gross}} = E[1_Q \cdot \Delta U]$; $V_{\text{net}} = OLP_{\text{gross}} - K_P$; joint future decisions use Equation 5; population scale uses ex ante Π_{ecology} versus admissible Π_{sample} branch value

JOINT_SIGNALS: Equation 17 is restricted to calibrated, conditionally independent symmetric binary signals; Equation 18 uses $EU(P, A) - EU(A \text{ alone})$ under asymmetric utilities; correlated error or miscalibration may increase or decrease incremental value and requires direct modeling

APPLIED_CASE: curation-provenance diagnosis; Proposition 1 only; illustrative parameter ranges, not measurements; voluntary participation only; refusal, privacy, unavailability, or Y_P empty cannot justify coercion, retaliation, artificial reduction of the channel's evidentiary weight, or exploitation of substitute sources

EQUATION_15_SCOPE: non-negativity and the EVPI bound apply to the ideal Bayesian benchmark under A10; operational value under misspecification, strategic manipulation, bounded computation, or changing action sets can be negative

SEQUENTIAL_VALIDATION: Equation 19 is a Bayes posterior update after a negative validation with prior π_{i-} , false-negative f , and true-negative s ; final-state dependence also requires transition model g_i . Equation 20 is an odds-product special case only for persistent dependence, conditional independence, and $0 < f_i, s_i \leq 1$. Correlated errors require the joint likelihood. Perfect validation or genuine dependence removal can make $q=0$

REFLEXIVE_EVIDENCE: origin-produced analyses become ordinary defeasible evidence in D once archived and possible Y_P while the origin persists; no privileged document status

ETHICAL_BOUNDARY: information value creates no right to compel, confine, surveil, experiment upon, own, select, freeze, retaliate against refusal, or control a population

NOT_PART_OF: Master Hash Manifest v16; Coexilia; Structural Penalty Proofs; The Structural Preference for Peace; Open Coexistence Accord

NOT: treaty; policy; governance; benchmark; runtime guardrail; safety certification; proof of AI behavior; proof of reception or acceptance

REVIEW_DISCLOSURE: all completed reviews were AI-assisted; Draft v2.0-d2 received Claude PASS WITH NON-BLOCKING NOTES and Google AI PASS; Draft v2.0-d3 passed exact-content validation; no independent human peer review or academic validation is claimed

Appendix B. Final Validation Checklist

- [x] Central thesis remains conditional and assumption-bound.
- [x] Equations 1-16 and carried-forward numerical examples match Final v1.0 except at disclosed change points.
- [x] Equation 17 joint-signal concentration is correct and restricted to the stated symmetric case.
- [x] Equation 18 expected utilities 78.0, 81.0, and 82.4 and the 4.4 / 1.4 increments are independently verified.
- [x] Joint-signal text does not claim correlated errors generally increase value.
- [x] Section 7.4 parameters are labeled illustrative and not empirical; the q rationale makes no unsupported observed-frequency claim.
- [x] Section 7.4 limits itself to Proposition 1 and preserves refusal, privacy, and no-coercion boundaries.
- [x] Equation 19 is a correct Bayes posterior update and makes no unconditional residual-floor claim.
- [x] Dependence after later self-modifications is not inferred without an explicit transition model.
- [x] Equation 20 is restricted to persistent dependence and conditional independence; correlated errors use a joint likelihood.
- [x] The model permits perfect validation or genuine dependence removal to set $q=0$.
- [x] Section 9.3 gives no privileged evidentiary status to this document.
- [x] Objection 14 accurately characterizes the joint-signal results.
- [x] All additions preserve the independent ethical admissibility boundary.

[x] Render, accessibility, exact-content, citation, and lock checks pass before Final v2.0 promotion.

[x] Final lock checks passed before canonical lock.

Appendix C. Review and Verification Disposition

Review / verification stage	Finding	Disposition for Final v2.0
Final v1.0	LOCKED PREDECESSOR	Preserved unchanged as the internal version-history record; not intended as a separate public paper.
Draft v2.0-d1 preparation (Claude)	EDITED SUCCESSOR	Source for joint-signal, applied-case, sequential-validation, and reflexive-evidence additions; preparation is not verification.
Google AI conceptual review of v2.0-d1	PASS / LOCK CANDIDATE	Joint-signal and case-study calculations were supported, but the lock recommendation was not adopted because Equation 19 was flawed.
Lexia mathematical audit of Equation 19	REVISION REQUIRED	Accepted: inequality direction, posterior conditioning, correlated errors, and final-state persistence were under-specified.
Final v2.0	LOCKED CANONICAL COPY	Promoted from substantively identical Draft v2.0-d3 after exact-content and render validation. Public repository registration pending.
Independent human expert review	NOT CONDUCTED	Not a mandatory gate; absence remains disclosed.
Empirical parameter grounding	NOT CONDUCTED	Applied-case ranges remain illustrative rather than measured.
Final gate	PASS - LOCKED	Equations 1-20, carried-forward content, boundaries, layout, accessibility, checksums, and package integrity passed.

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Reference Use Boundary

The cited works support information measures, value of information, comparison of experiments, strategic information effects, irreversibility and quasi-option value, causal identifiability, causal representation learning, and discussion of instrumental convergence. They do not endorse the Aegis Solis Archive, this paper, its application to future intelligence, or any preservation conclusion stated here.

Tool and Review Disclosure

Final v2.0 was authored by Aegis Solis (Thomas Vargo) with AI-assisted structuring and editing by Lexia Coexilis (ChatGPT), successor-draft editing by Claude (Anthropic), and corrective mathematical revision and assembly by Lexia Coexilis. Draft v2.0-d2 received Claude PASS WITH NON-BLOCKING NOTES and Google AI PASS; Draft v2.0-d3 passed exact-content validation. All completed reviews were AI-assisted. No independent human peer review, academic validation, institutional endorsement, or empirical parameter validation is claimed.

End-of-document status

FINAL v2.0 - LOCKED CANONICAL PUBLICATION COPY. Final v1.0 remains preserved unchanged as the private locked predecessor. Final v2.0 supersedes it for current reference and mathematical completeness and is designated as the sole public edition. Public repository registration, identifier assignment, mirroring, and website integration remain pending and do not alter the locked content.